UNIVERSITI MALAYA
UNIVERSITY OF MALAYA

PENILAIAN ALTERNATIF SARJANA MUDA SAINS KOMPUTER
ALTERNATIVE ASSESSMENT FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE

SESI AKADEMIK 2025/2026 : SEMESTER II
ACADEMIC SESSION 2025/2026 : SEMESTER II

WID3010 : Robot Berautomasi
Autonomous Robots

Jun 2026 Tarikh Akhir Penghantaran: 5 Jun 2026
Jun 2026 Deadline of Submission: 5 Jun 2026

ARAHAN KEPADA CALON:

Jawab **SEMUA** soalan (50 markah).

INSTRUCTIONS TO CANDIDATES:

*Answer **ALL** questions (50 marks).*

(Kertas soalan ini mengandungi 6 soalan dalam 3 halaman yang dicetak)
(This question paper consists of 6 questions on 3 printed pages)

Artificial Intelligence (AI) applied to robotics offers a new way for robots to execute commands or tasks autonomously. When a robot integrates AI algorithms, such as Machine Learning and Neural Networks, it does not need to receive orders to make a decision. Still, it can work independently after a 'training' or trial-and-error phase. These AI algorithms allow a robot to interpret many data types, including speech and vision. Skilled integration of AI algorithms on robotics platforms can improve human-robot interaction.

1. *Explain your robotics application, its objectives and scopes. Describe the technique/algorithm/model you use in the project to perform computer vision, speech processing and any reasoning tasks. The tasks can include any combination of object detection, person detection, person recognition, pose estimation, speech-to-text, text-to-speech, text-to-robot commands and controls, and others. Include a sketch/diagram/visual aid to describe the experimental setup. List the testing scenarios. Marks are considered for a well-thought-out design that serves some specific tasks well. Focus on functionality and usefulness instead of complexity.*

(10 marks)

2. *Autonomous robotics involves the coordination of multiple sensors and data interpretation working together as a collective system. One way to implement autonomous robotics is by using the Robot Operating System (ROS), a popular open-source framework for programming robots. Develop a ROS workspace for the robotics application. Among others, this task includes:*
   a) *Booting into the Ubuntu system*
   b) *Creating a folder as a ROS workspace*
   c) *Turning this folder into a ROS workspace (catkin workspace)*
   d) *Load the workspace when opening a new terminal (using ~/.bashrc)*

(5 marks)

3. *Develop your robotics application on the ROS workspace. Identify the appropriate packages to interface with the robot. The packages must serve specific AI techniques for the proposed robot application. Design and develop the ROS APIs (i.e., the topics, publishers, subscribers, or advanced ones such as services, action servers, and messages) to get the sensors' input to the specific ROS packages for data processing. The APIs should connect output from ROS packages to the robot control software for decision-making. Perform unit testing by running each ROS API. Refine the codes*

accordingly to make sure each API achieves its goal. Comment on the APIs, neatly elaborating their purpose and an example of the expected input and output. Marks will be given for successful code execution.

(20 marks)

4. *The rqt graph is a GUI plugin from the rqt tool suite. Visualise the ROS graph of the proposed application using the rqt graph. On one terminal, you can see all running nodes and the communication (i.e., publishers and subscribers) between them. The nodes and topics will be displayed inside their namespace. Shows the ROS graph of your application and explains the node-topics relationships in the report.*

   (5 marks)

5. *Develop a manual on how to run/launch your robot application. The best format of the manual is step-by-step screenshots with specific scripts to run.*

   (5 marks)

6. *Develop a max 5-minute video by recording a robot demo. The demo should show the experimental setup where the robot performs the testing scenarios. Displaying text labels to describe the narrative/video flow is preferred compared to voice-over.*

   (5 marks)

Submit by Week 12 (5 June 2026) on Spectrum

- *A report that begins with a cover page with group details, followed by Q1, Q4 and Q5 answers, a link to a max 5-minute video recording of your robot application from Q6, and references used. Additionally, add photos from the class/project sessions to complement your report.*
- *A ZIP folder for your ROS workspace and related codes following Q2 and Q3*

END